A-A-50967
18 April 1984

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, BENCH, HAND-POWER

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification 00-G-00661b.

This description covers hand operated bench mounted grinding machines used for sharpening edged hand tools. The grinding machine is intended to be used where it is impractical to provide power driven grinding machines.

Salient characteristics.

Classification. The grinding machines covered by this description shall be of the following types and of the sizes listed in Table I for the type machine required. The type and size machine furnished for a specific procurement shall be as specified in the contract.

Type I - Heavy duty, with at least 4 bearings.

Type II - Light duty, with at least 3 bearings.

Types and sizes. All types and sizes of grinding machines shall meet the requirements of Table I.

TABLE I. Sizes and capacities.

Type I - Heavy duty			Type II - Light duty			
Size	Size 6	Size 7	Size 4	Size 5	Size 6	Size 7
Wheel dimensions (inches)	6x1x3/8	7x1x3/8	4x7/8x5/16	5x1x3/8	6x1x3/8	7x1x3/8
Rev. of wheel to rev. of handle, not less than	10 to 1	12 to 1	9 to 1		10 to 1	

Supersession information. Due to the current description of bench grinding machines the term "medium duty" covered in Federal Specification 00-G-00661b has been superseded in this CID by the term "light duty." The type and size of Federal Specification 00-G-00661b and this CID are cross referenced in Table III.

FSC 3415

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Bearings. Type I grinding machines (heavy duty) shall have not less than four bearings. Two of the bearings shall support the grinding wheel shaft and two shall support the hand crank shaft. Type II grinding machines (light duty) shall have not less than three bearings. Two of the bearings shall support the grinding wheel shaft and one shall support the hand crank shaft.

Gear case. The grinding machine gear case shall be the same as that provided on the manufacturers' standard product and shall have the proper size and strength to support the grinding machine components. All encased gears shall operate smoothly and without excessive noise.

Tool rest. A tool rest shall be furnished and shall be adjustable to various heights and positions to accommodate numerous types of edged hand tools.

Hand crank. A hand crank and handle of the proper size and length shall be furnished to impart rotary motion to the grinding wheel.

Grinding wheel. A grinding wheel of the size specified in Table I shall be furnished with each machine. The wheel shall have aluminum oxide abrasives, vitrified bond, medium grain, medium grade of hardness, square faced and straight sides and shall meet the requirements of ANSI B74.13.

Mounting flanges. Mounting flanges shall be provided to prevent slippage of the grinding wheel during operation.

Screw clamp. A screw clamp shall be furnished with Type I machines and shall accommodate bench tops 1 to 3 inches in thickness. A screw clamp shall be furnished with Type II machines which shall accommodate bench tops 3/4 to 1-1/2 inches thick.

Lubrication. A suitable means of lubrication shall be provided to lubricate all bearings and gears. Hinge lid oil caps shall be provided on all external oil holes to prevent grit or dust from entering.

Paint. External surfaces shall be painted in accordance with the manufacturers' standard practice.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this CID where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Quality conformance inspection. The supplier shall randomly select the number of samples specified in Table II that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with all requirements specified herein. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of Table II, the requirements specified herein and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of Table II, the entire lot shall be considered defective and unsuitable for acceptance.

TABLE II. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance number defective	Rejection number defective
40 and under	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 to 800	25	2	3
801 to 1300	35	3	4
1301 and over	50	4	5

Testing. Sample grinding machines selected in accordance with Table II shall be subjected to the following tests:

Screw clamp test. The grinding machine shall be fastened to a bench top in the usual manner. The screw clamp shall be subjected to a force or torsional moment of 120 inch-pounds using a torque wrench, spring balance or other load-measuring device. The force shall be applied at right angles to the longitudinal axis of the screw. No parts of the adjusting screw or clamp shall bend, break, or show any visual indications of weakness.

Crank test. Clamp the driving gear shaft of the grinding machine so that it remains stationary. Using a spring balance, apply 150 inch-pounds of force to the crank handle in the clockwise direction of rotation. Mark the movement of the crank handle from the stationary position. Apply 150 inch-pounds of force to the crank handle in the counterclockwise direction of rotation and mark the

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movement as above. The difference in the clockwise and counterwise movement of the crank handle shall not exceed 1/4 inch. A force of 100 inch-pounds shall be applied to the end of the crank handle in a direction tending to pull the crank away from the gear case. The crank handle shall not break or show permanent deformation when so tested.

Operational test. The grinding machine shall be subjected to a practical grinding test of sufficent duration to determine whether the operation is easy and smooth and entirely satisfactory for the purpose intended. There shall be no wheel slippage allowed during the grinding operations.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Measurement system. In this description, all measurements, dimensions, sizes and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be in accordance with ASTM Standard D 3951, Commercial Packaging, or as specified in the contract or order.

Notes. American National Standards Institute ANSI B74.13, "Markings for Identifying Grinding Wheels and Other Bonded Abrasives", and ANSI/IEEE 268, "Metric Practice" is available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

ASTM-D-3951 - "Commercial Packaging, Standard Practice for", is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Ordering data. The purchaser will specify quantity, type, and size grinder required. In fulfillment of the contract the contractor shall supply the quantity, types, and sizes specified in the contract or order.

TABLE III. Cross reference.

00-G-00661 Heavy duty size	A-A-50967 Heavy duty size	00-G-00661 Medium duty size	A-A-50967 Light duty size
5	-	4	4
6	6	5	5
7	7	6	6
-	-	7	7

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MILITARY INTERESTS:

Custodian

Air Force - 84

Review Activities

DLA - GS, DH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3415-0149

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